# The Perks of Being a First Mover

Does White's Advantage in Chess Change with Rating, Opening and Time Control?

CS215 Open Project · Final Report · Group 02

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https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2

## Summary

We measure the first-move advantage on Lichess using 162,175,412 standard rated games from 2024-01 and 2024-02. Instead of comparing raw win rates, we fit White's advantage as a number of *rating points*, $h$, after correcting every game for the rating gap between the players. Overall White is worth $h = +11.9 \pm 0.1$ rating points. We then ask three questions: how $h$ changes with playing strength, with the time control (Bullet to Classical), and with the opening, where we compare four openings with an aggressive reputation (all 1.e4 gambits) against four with a drawish reputation. **[one or two sentences with the main findings, written once the real results are in]**

## 1. Question and hypotheses

Chess is a game of perfect information in which White moves first. Database statistics have long shown that White scores a little better than Black, but a raw win rate mixes three things: the first move itself, the rating gap between the players, and how often games are drawn. We separate them and test:

**H1 Rating (from our proposal).** White's advantage is not constant across skill levels. It peaks at intermediate ratings, where players can exploit the initiative but defend poorly, and shrinks at 2200+, where Black knows how to neutralise it.

**H2 Openings (new for the final stage).** Aggressive openings, where White gives up material for the initiative, carry a larger White advantage than drawish openings, and drawish openings have far more draws at every rating.

**H3 Time control.** White's advantage depends on how much time the players have. In slower games the first-move initiative is used more accurately and fewer games are decided by the clock, so we expect $h$ to grow from Bullet to Classical.

## 2. Data

**Source.** The Lichess Open Database (`https://database.lichess.org`), standard rated games for 2024-01 and 2024-02, released under the Creative Commons CC0 licence. Each month is one zstd-compressed PGN file of tens of GB. Every game has about 17 header tags. We use seven: `Result`, `WhiteElo`, `BlackElo`, `Event` (gives the speed), `TimeControl`, `Termination` and `Opening`.

| Month | Games read | Games kept | Kept |
|---|---|---|---|
| 2024-01 | 98,994,760 | 84,177,790 | 85.0% |
| 2024-02 | 91,567,975 | 77,997,622 | 85.2% |
| Total | 190,562,735 | 162,175,412 | 85.1% |

**Cleaning.** We drop games with no result (6,007), abandoned games (502,851), games with a missing rating (0), and games where the two ratings differ by more than 100 points (27,878,465). The rating gap is corrected for inside the model anyway, but very lopsided games say little about the first move. Each game is put in a 100-point rating bracket $[a, a{+}100)$ by the *average* of the two ratings, and in a speed (UltraBullet to Correspondence) from the `Event` tag.

**Openings.** Lichess names the opening of every game, for example *King's Gambit Accepted: Bishop's Gambit*. We match those names as shown in Table 1. Inside the drawish openings we leave out their gambit side-lines, such as the Stafford Gambit in the Petrov, so that the drawish group really is quiet. The eight openings cover 11,367,920 games (7.0% of all kept games). Everything else is "other".

**Table 1:** Openings and the Lichess opening names they cover. A game counts if its name starts with one of the listed names and contains none of the excluded ones.

| Opening | Group | Lichess name starts with | Excluding |
|---|---|---|---|
| King's Gambit | Aggressive | *King's Gambit* | – |
| Evans Gambit | Aggressive | *Italian Game: Evans Gambit* | – |
| Danish Gambit | Aggressive | *Danish Gambit* | – |
| Smith-Morra Gambit | Aggressive | *Sicilian Defense: Smith-Morra Gambit* | – |
| Petrov's Defense | Drawish | *Petrov's Defense*, *Russian Game* | Stafford Gambit, Cochrane Gambit |
| Berlin Defense | Drawish | *Ruy Lopez: Berlin Defense* | – |
| Queen's Gambit Declined | Drawish | *Queen's Gambit Declined* | Albin Countergambit, Chigorin Defense, Baltic Defense |
| French Exchange | Drawish | *French Defense: Exchange Variation* | – |

**Why these eight.** The aggressive openings are White gambits: White gives up a pawn or more for fast development and attack (King's, Evans, Danish and Smith-Morra Gambits). The drawish openings have a reputation for symmetrical, simplified positions: the Petrov and the Berlin (Black's choice), the Queen's Gambit Declined (Black's choice), and the French Exchange (White's choice to simplify).

## 3. Method

The full derivations, with worked examples, are in the separate document *Theory.pdf*.

### 3.1 White's advantage in rating points

Under the Elo model, a player rated $d$ points above the opponent is expected to score $E(d) = 1/(1+10^{-d/400})$. We give White an extra $h$ points:

$$E_i(h) = \frac{1}{1+10^{-(R_i^{\mathrm{W}}+h-R_i^{\mathrm{B}})/400}},$$

and choose the $h$ that makes the expected scores add up to the actual scores ($S_i = 1, \frac{1}{2}, 0$ for a White win, draw or loss). This is the maximum-likelihood estimate. We find it with Newton's method, and its standard error comes from the curvature of the log-likelihood (the Fisher information). Reporting $h$ in rating points makes groups comparable: a 1540 who beats a 1570 with White counts as stronger evidence than a win between two 1570s. Draws count as half a point. Draw and win rates are reported separately.

### 3.2 $h$ by rating (H1)

We fit $h$ separately in every 100-point rating bracket (by the average of the two ratings), for each speed, and read the shape from the chart of $h$ against rating. We don't fit a single line through the brackets, because the shape is not close to a straight line, and a slope would hide it.

### 3.3 Aggressive vs drawish (H2)

Each opening is treated as its own set of games: we fit $h$ by maximum likelihood using only the games played in that opening, which gives one $h$ (with its CI) per opening. For the comparison we also fit one $h$ for all four aggressive openings together and one for the four drawish ones,

and test the difference with a $z$-test: the two groups are different games, so the variances add, $\mathrm{SE}(\hat{h}_A - \hat{h}_D) = \sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}$. We do this for all speeds together and for each of Bullet, Blitz and Rapid.

### 3.4 Time controls (H3)

Lichess sorts every game into a speed (UltraBullet, Bullet, Blitz, Rapid, Classical, Correspondence) by its time control. We treat each speed exactly like an opening: its games are a separate set, with their own maximum-likelihood $h$ and win/draw/loss shares. We then compare each speed with the next faster one using the same $z$-test for a difference.

### 3.5 Big data and significance

With 162,175,412 games, almost any difference has a tiny $p$-value. So we always report the size of an effect with its 95% confidence interval, and treat effects of a few rating points as small whatever their $p$-value.

### 3.6 Tools

A C++ program streams the compressed files (zstd), reads only the header lines, and counts games by opening, speed, rating bracket and rating gap. The two months are read on two threads. We ran it on Google Colab, streaming the files straight from Lichess so nothing large was downloaded. It saves those counts (`counts.csv`, a few MB), and every table, test, figure, this report and an interactive Plotly dashboard (`output/final/dashboard.html`) are rebuilt from them in seconds.

## 4. Results

### 4.1 Overall

Over all kept games White wins 49.7%, draws 4.0% and loses 46.3%. White's average score is 0.517 against 0.500 predicted by the ratings alone, which is $h = +11.9 \pm 0.1$ rating points.

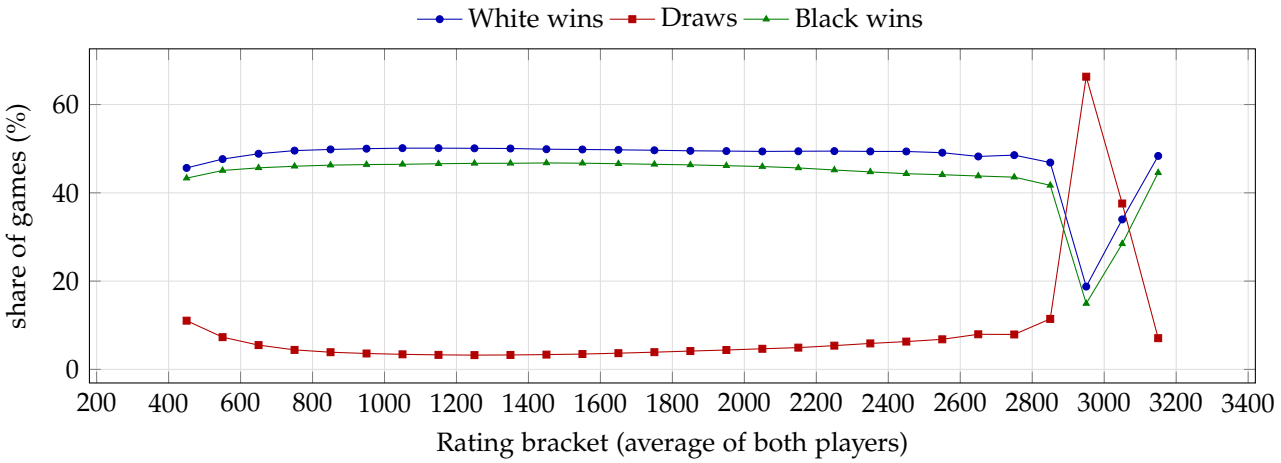



**Figure 1:** Share of games won by White, drawn and won by Black, by rating (all speeds).

### 4.2 Does the advantage peak at intermediate ratings? (H1)

**[interpretation: is H1 supported?]**

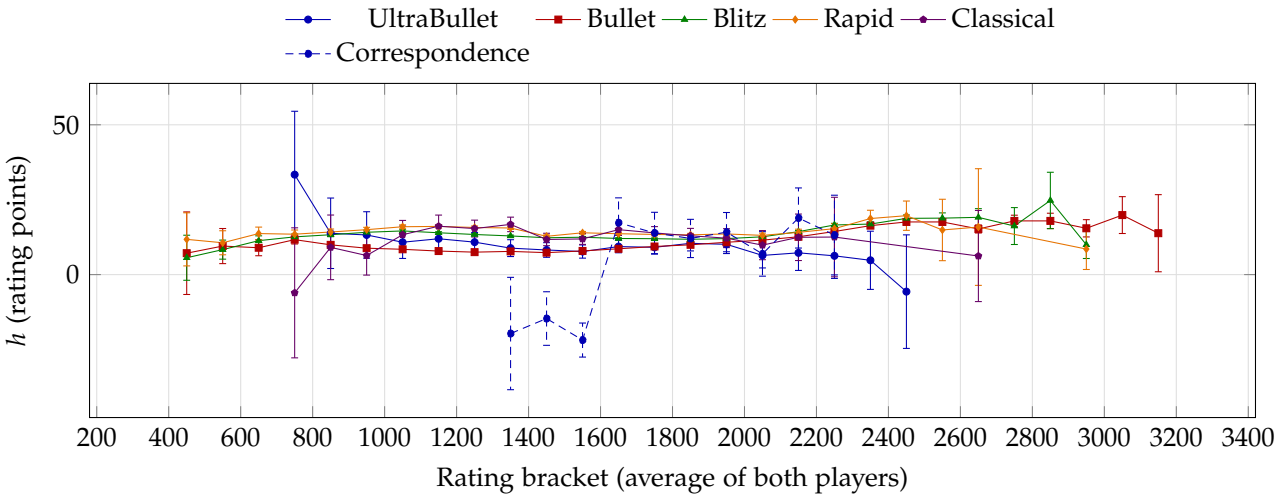



**Figure 2:** $h$ in each 100-point rating bracket, by speed, with 95% CIs. Brackets with fewer than 1,000 games are left out.

### 4.3 Openings (H2)

**Table 2:** White's advantage by opening (all speeds). Avg rating = average rating of the games.

| Opening | Games | Avg rating | White | Draw | Black | $h$ (pts) |
|---|---|---|---|---|---|---|
| King's Gambit | 2,282,379 | 1633 | 51.3% | 3.0% | 45.6% | $+18.4 \pm 0.5$ |
| Evans Gambit | 165,739 | 1680 | 51.9% | 3.3% | 44.8% | $+24.0 \pm 1.7$ |
| Danish Gambit | 484,805 | 1635 | 53.8% | 3.2% | 43.0% | $+37.1 \pm 1.0$ |
| Smith-Morra Gambit | 1,136,895 | 1727 | 49.6% | 3.6% | 46.8% | $+11.2 \pm 0.6$ |
| Petrov's Defense | 2,071,985 | 1460 | 50.4% | 3.9% | 45.7% | $+15.5 \pm 0.5$ |
| Berlin Defense | 579,944 | 1544 | 50.3% | 5.6% | 44.1% | $+20.8 \pm 0.9$ |
| Queen's Gambit Declined | 3,660,887 | 1729 | 51.3% | 4.5% | 44.2% | $+23.9 \pm 0.4$ |
| French Exchange | 985,286 | 1765 | 48.1% | 4.9% | 47.1% | $+4.2 \pm 0.7$ |
| Aggressive (all 4) | 4,069,818 | 1662 | 51.2% | 3.2% | 45.6% | $+18.8 \pm 0.3$ |
| Drawish (all 4) | 7,298,102 | 1643 | 50.5% | 4.4% | 45.0% | $+18.6 \pm 0.3$ |
| Other openings | 141,698,824 | 1642 | 49.5% | 4.0% | 46.5% | $+10.7 \pm 0.1$ |

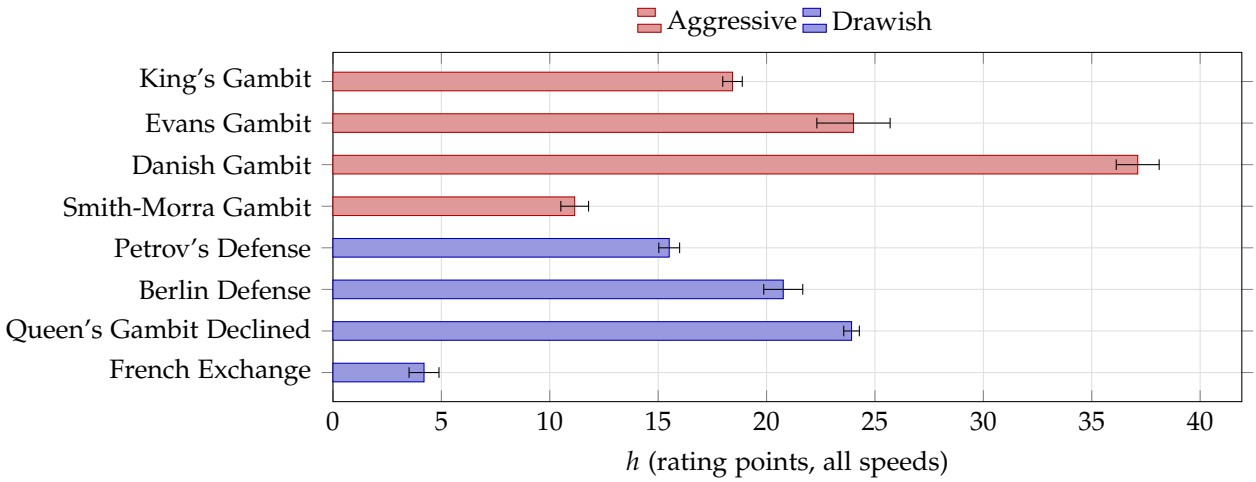



**Figure 3:** $h$ for each opening (all speeds) with 95% CIs.

Over all speeds, $h$ for the aggressive openings minus $h$ for the drawish openings is $+0.2 \pm 0.4$ rating points ($z = 1.0$, $p = 0.298$). **[interpretation: is H2 supported, and what does it say about White's**

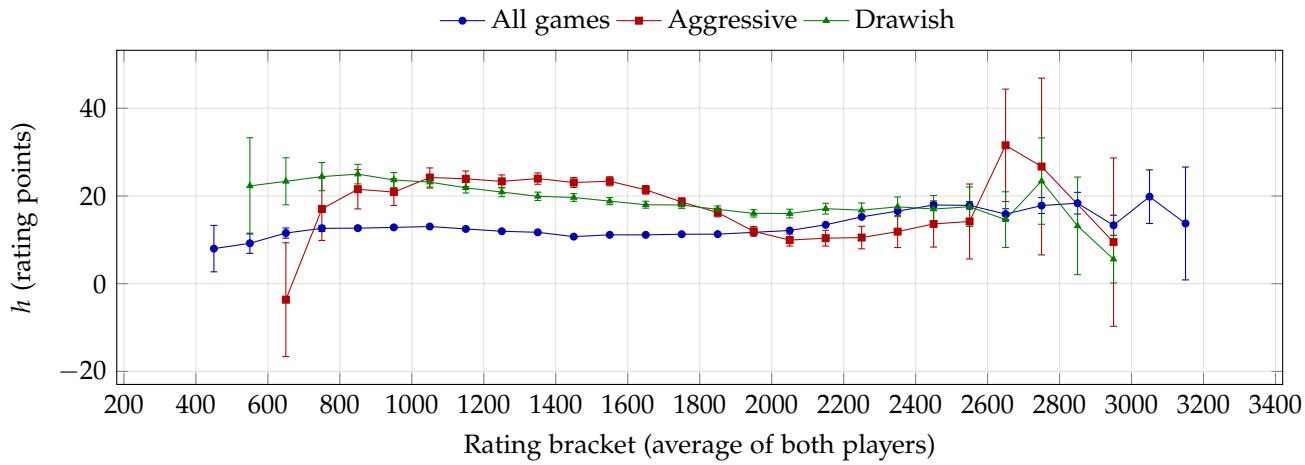



**Figure 4:** $h$ by rating for the aggressive and the drawish openings, with 95% CIs.

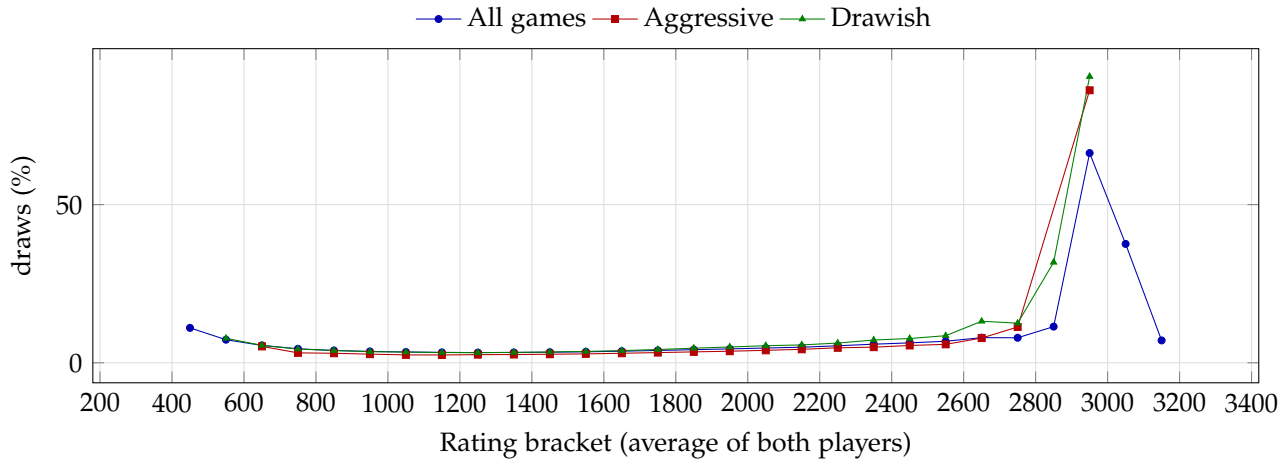



**Figure 5:** Draw rate by rating for the aggressive and the drawish openings.

**Table 3:** Aggressive vs drawish openings: $h$ for each group, and the difference with its $z$-test (rating points, 95% CI), for all speeds and for each of the main speeds.

| Speed | Aggressive | Drawish | Difference | $z$ | $p$ |
|---|---|---|---|---|---|
| All speeds | $+18.8 \pm 0.3$ | $+18.6 \pm 0.3$ | $+0.2 \pm 0.4$ | +1.0 | 0.298 |
| Bullet | $+16.9 \pm 0.6$ | $+16.3 \pm 0.4$ | $+0.6 \pm 0.8$ | +1.7 | 0.097 |
| Blitz | $+19.5 \pm 0.5$ | $+18.9 \pm 0.4$ | $+0.6 \pm 0.6$ | +2.0 | 0.044 |
| Rapid | $+20.3 \pm 0.9$ | $+22.4 \pm 0.6$ | $-2.1 \pm 1.1$ | -3.7 | $1.8 \times 10^{-4}$ |

**advantage?]**

### 4.4 Time controls (H3)

**Table 4:** White's advantage by speed (all openings). White / Draw / Black = share of games won by White, drawn, won by Black. $h$ with 95% CI.

| Speed | Games | Avg rating | White | Draw | Black | $h$ (pts) |
|---|---|---|---|---|---|---|
| All speeds | 162,175,412 | 1638 | 49.7% | 4.0% | 46.3% | $+11.9 \pm 0.1$ |
| UltraBullet | 722,976 | 1611 | 50.3% | 2.0% | 47.8% | $+9.1 \pm 0.8$ |
| Bullet | 59,945,392 | 1716 | 49.9% | 3.0% | 47.1% | $+9.9 \pm 0.1$ |
| Blitz | 77,918,475 | 1613 | 49.6% | 4.4% | 46.0% | $+12.8 \pm 0.1$ |
| Rapid | 22,650,621 | 1520 | 49.6% | 4.9% | 45.5% | $+14.2 \pm 0.1$ |
| Classical | 857,174 | 1569 | 49.1% | 5.7% | 45.2% | $+13.4 \pm 0.7$ |
| Correspondence | 80,774 | 1789 | 47.5% | 5.7% | 46.8% | $+2.8 \pm 2.4$ |

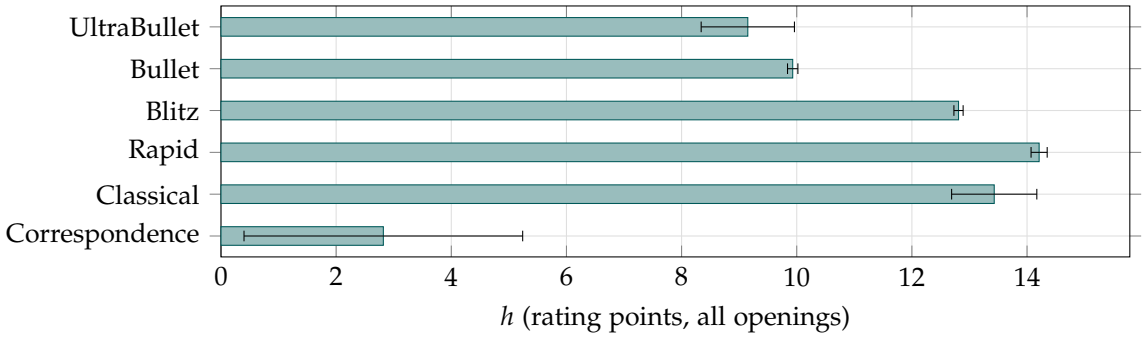



**Figure 6:** $h$ for each speed (all openings) with 95% CIs.

**Table 5:** Each speed against the next faster one: $h$ for both, and the difference (slower minus faster) with its $z$-test (rating points, 95% CI).

| Speeds | Slower | Faster | Difference | $z$ | $p$ |
|---|---|---|---|---|---|
| Bullet vs UltraBullet | $+9.9 \pm 0.1$ | $+9.1 \pm 0.8$ | $+0.8 \pm 0.8$ | +1.9 | 0.060 |
| Blitz vs Bullet | $+12.8 \pm 0.1$ | $+9.9 \pm 0.1$ | $+2.9 \pm 0.1$ | +47.8 | $< 10^{-300}$ |
| Rapid vs Blitz | $+14.2 \pm 0.1$ | $+12.8 \pm 0.1$ | $+1.4 \pm 0.2$ | +16.9 | $9.6 \times 10^{-64}$ |
| Classical vs Rapid | $+13.4 \pm 0.7$ | $+14.2 \pm 0.1$ | $-0.8 \pm 0.8$ | -2.0 | 0.042 |
| Correspondence vs Classical | $+2.8 \pm 2.4$ | $+13.4 \pm 0.7$ | $-10.6 \pm 2.5$ | -8.2 | $2.3 \times 10^{-16}$ |

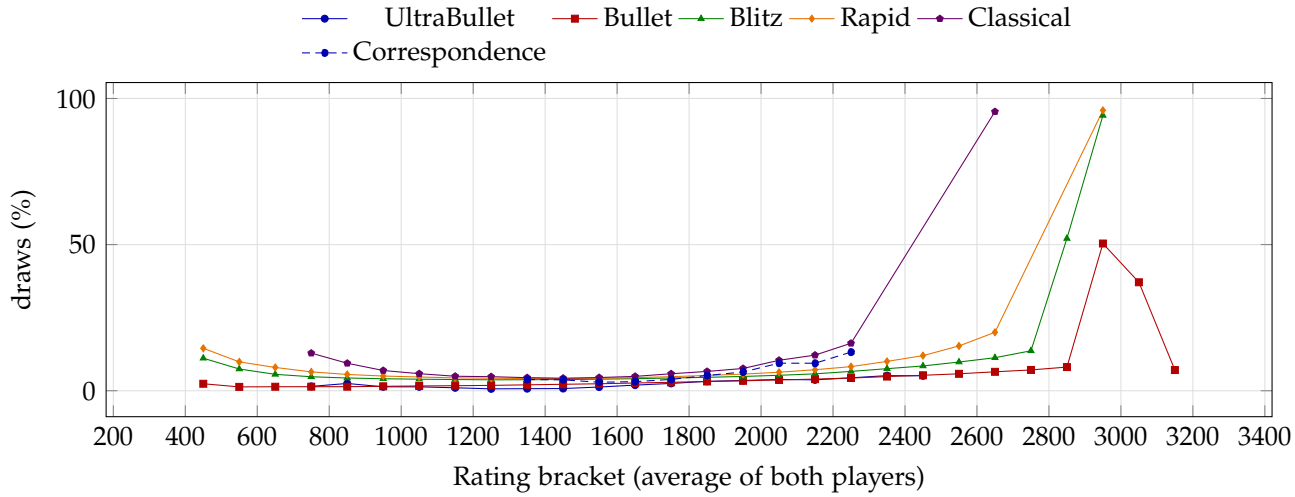



**Figure 7:** Draw rate by rating for each speed.

**[interpretation: is H3 supported?]**

## 5. Discussion

**[what the results mean for White's advantage, written once the real results are in]**

**Limitations.**

- *Choosing an opening is not random.* Players who play the King's Gambit or the Berlin differ in style and preparation, and the opening is chosen by one player (White for the gambits and the French Exchange, Black for the Petrov, Berlin and QGD). Our $h$ corrects for the rating gap in each game, but the results show association, not that the opening causes the difference.
- *The groups have different players.* The comparison of the two opening groups doesn't adjust for their players' ratings. The "Avg rating" column and Figure 4 (which compares the groups bracket by bracket) show how much that matters.
- *Ratings are noisy*, especially for new accounts, which blurs the correction for the rating gap. Lichess ratings also differ from FIDE ratings, so "2200 on Lichess" is not a master.
- *Games are not independent.* Active players contribute thousands of games, so the real uncertainty is somewhat larger than our confidence intervals.
- *One site, two months.* The results describe Lichess players in early 2024.

## 6. Conclusion

**[answer to the three questions in a few sentences]**

### Contributions

| | |
|---|---|
| Sanidhya | Statistical modelling: fitting $h$, confidence intervals and the $z$-tests. |
| Daksh Maahor | Data cleaning and opening classification; win/draw/loss rates. |
| Sahil | Data engineering: the C++ streaming parser and the Colab pipeline. |
| VadaPavJai | Visualisation: figures, the Plotly dashboard and the report. |

### References

Lichess Open Database, `https://database.lichess.org`. Lichess opening names, `https://github.com/lichess-org/chess-openings`. A. E. Elo, *The Rating of Chessplayers, Past and Present*, Arco, 1978. M. E. Glickman, "Example of the Glicko-2 system", 2012.